



SENTIENT TOKEN AND ECONOMIC OPTIMIZATION FOR AGENTS

Make agent optimization falsifiable.

An open operation-control and evidence layer that binds what an agent was asked to do, what actually ran, what it cost, and whether an external verifier accepted the result.

PUBLIC-GOODS REQUEST

\$150,000

Nine months. Three go/no-go tranches. No equity. Open methods, code, evidence, and negative results.

808

public GitHub stars

168

signed prospective cells

0

adversarial false passes

OPEN

MIT code and public proof

PROBLEM

The model call is not the economic unit.

Real cost and failure accumulate across the entire operation.

- Decomposition, parallelism, retries, tools, context, recovery, and verification can overwhelm a cheap model choice.
- A requested model can differ from the provider path that actually executes.
- Unknown local, subscription, setup, or human cost is often collapsed into zero.
- Model prose or stack status can be promoted to completion without an external artifact check.

01 Declared plan

Models, tools, topology, and limits

02 Observed execution

Calls, attempts, artifacts, and cost lenses

03 External verdict

Deterministic verifier outside the route

04 Signed evidence

Offline-reconstructable receipt chain

UNKNOWN STAYS UNKNOWN



WORKING SUBSTRATE

Built before the grant.

Citadel starts with one command, then exposes stronger control only when the operation needs it.

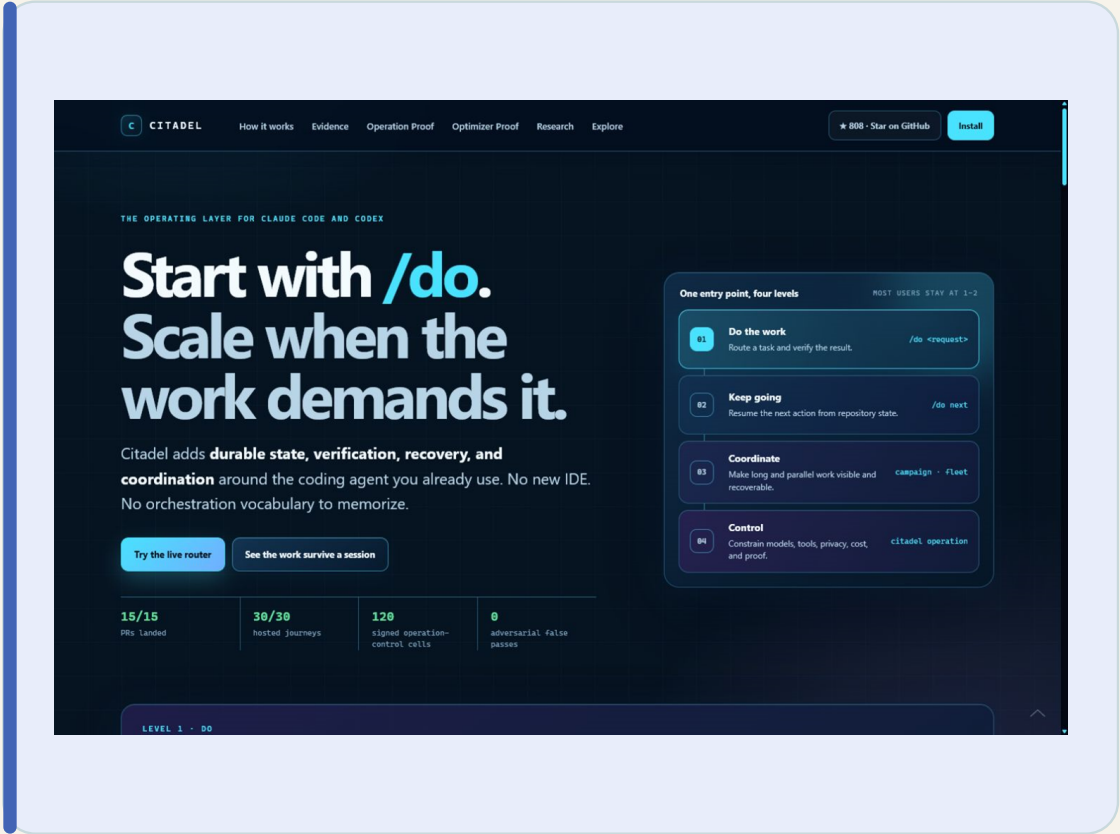
- /do

Beginner path
Natural-language routing, verification, and a durable next action.
- PLAN

Portable contract
Models, tools, topology, retries, privacy, time, cost, and stop rules.
- RUN

Runtime coverage
Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- PROOF

Evidence plane
External artifact verdicts, failure preservation, signed receipts, and offline replay.





PINNED SENTIENT ROMA DIAGNOSTIC

The proof passed. The optimizer gate did not.

A frozen 24-cell run compared frontier, prompt routing, direct open/local 7B, and Citadel-controlled ROMA.

POLICY	VERIFIED	DURATION	FRONTIER	LOCAL / MODULE
Frontier-only	6 / 6	47.7s	6	0
Prompt router	3 / 6	40.8s	3	3
Direct open/local 7B	2 / 6	13.2s	0	6
Citadel + ROMA	4 / 6	1042.8s	0	89 module calls

24 / 24

measured cells

0

false passes

4 / 6

Citadel local verified

FAILED

strong-attempt savings gate



NEGATIVE EVIDENCE IS A DELIVERABLE

Citadel rejected its own easy story.

Three frozen studies changed the plan instead of being marketed into a savings claim.

V1 - TIMEOUT SENSITIVITY

NO SAVINGS CLAIM

27/36 vs 24/36 verified. Apparent -9.9% energy reversed to +3.5% energy and +5.4% modeled cost after excluding one matched timeout pair.

V2 - CAPABILITY PROFILE

ECONOMIC REGRESSION

Equal 24/36 completion, but 12 verifier escalations produced +15.7% measured GPU energy and +16.4% modeled cost.

V3 - REPOSITORY ARTIFACTS

FROZEN GATE MISSED

Both policies verified 6/12 cells. A 7.1% energy reduction missed the frozen 20% gate; token use increased 13.2%.



NINE-MONTH RESEARCH AND ENGINEERING PROGRAM

Funding buys the unproven result.

Scale a working control and proof layer into a representative optimizer - or publish a reusable negative result.

01	Representative workload	60+ artifact-producing operations and complete cost boundary	\$30k
02	Adapter SDK	Sentient ROMA plus two additional open stacks	\$28k
03	Expected-value controller	Price verification, escalation, recovery, and failure	\$35k
04	Prospective evaluation	Frozen multi-stack policy and statistical contract	\$32k
05	Public proof and release	Hosted reconstruction, /do docs, cohort, and maintenance	\$25k



NO-GO RULES ARE PART OF THE PRODUCT

Clear spend. Hard outcome gates.

COST BASIS

Maintainer labor	\$99,000
Compute and tools	\$27,000
Hosted reproducibility	\$7,000
Docs and accessibility	\$7,000
Operator cohort	\$2,000
Bounded contingency	\$8,000

Total **\$150,000**

PRIMARY PROSPECTIVE GATE

> = 80%

absolute verified completion

> = 95%

of valid frontier rate

> = 30%

lower measured end-to-end cost

ZERO

false passes and integrity failures

UNKNOWN COST CANNOT COUNT AS SAVINGS

DIRECT FIT WITH A NAMED REQUEST FOR PRODUCTS

An open optimizer that can prove when it is wrong.

WHY FUND CITADEL

- Directly targets Sentient's Token and Economic Optimization request.
- Already binds a pinned Sentient ROMA stack without replacing its planner.
- Signed evidence lowers substrate risk and exposes the remaining research risk.

WHAT STAYS PUBLIC

- MIT code, adapters, contracts, methods, verifiers, cells, and reports.
- No mandatory Citadel account, token, hosted evidence endpoint, or telemetry resale.
- Users keep project state, operations, receipts, and the right to run locally.

Seth Gammon - solo builder and maintainer

Evaluator: github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR_START_HERE.md

OPEN. VERIFIABLE. FALSIFIABLE.